

DYLAN LOUW

BSc Information Technology Student | 062 005 9562 | djlouw2005@gmail.com

LinkedIn: [Dylan Louw](#) | GitHub: [Dylan Louw](#) | Portfolio Website: [My Portfolio](#)

ABOUT ME

I am a final-year IT student at North-West University who brings a strong logical mindset to software development. I enjoy structuring clean code, designing backend logic, and building reliable systems. My skills are backed by solid project results, including a 90% mark for a 2nd-year C#/SQL project and an 84% mark for a 3rd-year database design. I am bilingual (English and Afrikaans) and am finishing my undergraduate degree with plans to pursue my Honours in BSc IT and Computer Science in 2027.

TECHNICAL SKILLS & CERTIFICATIONS

- **Trained in** C#, Java (OOP), C++, Python, Delphi (Object-Pascal), SQL.
- **Working knowledge of** core data structures (Linked Lists, Stacks, Trees) and basic algorithm complexity (Big-O notation).
- **Systems Analysis & Design:** Exposure to the Software Development Life Cycle (SDLC), Requirements Analysis, and UML Modeling.
- **Web Development:** Foundational HTML and CSS (Self-Taught).
- **Cybersecurity:** Certified in Introduction to Cybersecurity (Great Learning) and Information Security (NWU-CMPG215).

EDUCATION

Hoërskool Noordheuwel | Krugersdorp | 2019 – 2023 National Senior Certificate (NSC) with 2 Distinctions

- **77% for Information Technology** (Delphi foundation).
- **83% for Computer Applications Technology (CAT).**

North-West University (NWU) | Potchefstroom | 2024 – 2026 (Expected) Bachelor of Science in Information Technology

- **Academic Progress:** Completed **328 of 392 credits** (NQF Level 7).
- **Key Distinctions:**
 - Structured Programming (CMPG121).
 - Financial Accounting (ACCS121).
 - Communication Skills (CMPG214).
 - Understanding the Natural World (WVNS211).
 - Databases (CMPG311).
- **Technical Resilience:** Achieved 69% proficiency in *Object Oriented Programming* and *System Analysis and Design 2* following targeted improvement. Additionally passed *User Interface Programming (CMPG212)* at 73%, *Decision Support Systems (CMPG312)* at 64%, *Artificial Intelligence (CMPG313)* at 64%, and *Computer Networks (CMPG315)* at 66%.

FEATURED PROJECTS

Mobile Pothole Reporting & Dispatch System (2nd Year) [\[GitHub\]](#)

- **Result:** 90% (Group Project)
- **Architecture & Design:** Co-designed system architecture and UML diagrams to map out real-time interactions between users and municipal dispatchers.
- **Full-Stack Development:** Developed backend logic using C# and integrated SQL databases to handle location-based reporting and data persistence.
- **System Logic:** Applied robust conditional and Boolean logic to handle report severity algorithms and manage secure user access permissions.

Corporate Database Design & Implementation (3rd Year)

- **Result:** 84% (Group Project)
- **Database Architecture:** Analysed complex business rules to design precise Entity-Relationship Diagrams (ERDs), ensuring data integrity and optimal normalization before writing code.
- **Implementation:** Translated logical designs into a fully functional relational schema using Oracle SQL, writing clean queries and structuring constraints effectively.

Personal Portfolio Website (Personal Project) [\[GitHub\]](#) [\[Live Site\]](#)

- **Frontend Development:** Designed and built a fully responsive personal portfolio website from scratch using HTML, CSS, and JavaScript, structured across multiple sections including About, Experience, Projects, and Contact.
- **UI/UX & Animation:** Implemented a custom dark-mode theme using CSS variables for maintainable theming, alongside scroll-triggered fade-in animations built with the Intersection Observer API and interactive hover effects across navigation and project cards.
- **Responsive Engineering:** Built a mobile hamburger navigation system with custom JavaScript toggle logic and CSS media queries to ensure consistent usability across desktop, tablet, and mobile devices.

CORE COMPETENCIES

- **Analytical Logic & Data Structures:** Strong foundation in Discrete Mathematics and Graph Theory, applying these concepts to solve complex programmatic problems and optimize algorithmic logic.
- **Architectural Precision:** A critical approach to system design, focusing on identifying potential risks, debugging backend logic, and ensuring code integrity.
- **Technical Adaptability:** Highly adaptable to new languages and frameworks by focusing on foundational programming theory and core architectural principles.

LANGUAGES

- **Afrikaans:** Native / Home Language
- **English: Full Professional Proficiency** (74% in Matric & completed NWU Academic Literacy / ALDE112 & ALDE122)